

Задание часть 3. Встраиваемые системы
Задание 17 - по ядру Linux (кросскомпиляция)

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- 1) Поставил кросскомпилятор

```
komal@komal-VirtualBox:~$ sudo apt install gcc-arm-linux-gnueabihf
Installing:
gcc-arm-linux-gnueabihf
```

- 2) Создал конфиг на базе arm

```
CLEAN arch/arm/include/generated
komal@komal-VirtualBox:~/linux$ ARCH=arm make CROSS_COMPILE=arm-linux-gnueabihf- multi_v7_defconfig
HOSTCC scripts/basic/fixdep
HOSTCC scripts/kconfig/conf.o
```

- 3) Далее собираю ядро

```
komal@komal-VirtualBox:~/linux$ ARCH=arm CROSS_COMPILE=arm-linux-gnueabihf- make zImage
```

```

KSYMS .tmp_vmlinux0.kallsyms.S
AS .tmp_vmlinux0.kallsyms.o
LD .tmp_vmlinux1
NM .tmp_vmlinux1.syms
KSYMS .tmp_vmlinux1.kallsyms.S
AS .tmp_vmlinux1.kallsyms.o
LD .tmp_vmlinux2
NM .tmp_vmlinux2.syms
KSYMS .tmp_vmlinux2.kallsyms.S
AS .tmp_vmlinux2.kallsyms.o
LD vmlinux.unstripped
NM System.map
SORTTAB vmlinux.unstripped
OBJCOPY vmlinux
GEN modules.builtin.modinfo
GEN modules.builtin
OBJCOPY arch/arm/boot/Image
Kernel: arch/arm/boot/Image is ready
LDS arch/arm/boot/compressed/vmlinux.lds
AS arch/arm/boot/compressed/head.o
GZIP arch/arm/boot/compressed/piggy_data
AS arch/arm/boot/compressed/piggy.o
CC arch/arm/boot/compressed/misc.o
CC arch/arm/boot/compressed/decompress.o
CC arch/arm/boot/compressed/string.o
AS arch/arm/boot/compressed/hyp-stub.o
CC arch/arm/boot/compressed/fdt_rw.o
CC arch/arm/boot/compressed/fdt_ro.o
CC arch/arm/boot/compressed/fdt_wip.o
CC arch/arm/boot/compressed/fdt.o
CC arch/arm/boot/compressed/atags_to_fdt.o
CC arch/arm/boot/compressed/fdt_check_mem_start.o
AS arch/arm/boot/compressed/lib1funcs.o
AS arch/arm/boot/compressed/ashldi3.o
AS arch/arm/boot/compressed/bswapsdi2.o
LD arch/arm/boot/compressed/vmlinux
OBJCOPY arch/arm/boot/zImage
Kernel: arch/arm/boot/zImage is ready
komal@komal-VirtualBox:~/linux$

```

- 4) Скопировал в свою папку

```

komal@komal-VirtualBox:~/linux/arch/arm/boot$ cp zImage ~/Task_18/zImage

```

- 5) Превращаем все dts в dtb бинарные файлы

```

komal@komal-VirtualBox: ~/linux
komal@komal-VirtualBox:~/linux$ ARCH=arm make dtbs

```

- 6) Скопировал dtb файл к себе в папку

```

arm-realview-eb-bbrevid.dts      arm-realview-pbx-a9.dts      mps2-an399.dts      vexpress-v2p-ca
komal@komal-VirtualBox:~/linux/arch/arm/boot/dts/arm$ cp vexpress-v2p-ca9.dtb ~/Task_18/vexpress-v2p-ca9.dtb
komal@komal-VirtualBox:~/linux/arch/arm/boot/dts/arm$

```

- 7) Запускаю через qemu

```

komal@komal-VirtualBox:~/Task_18$ QEMU_AUDIO_DRV=none qemu-system-arm -M vexpress-a9 -kernel zImage -dtb vexpress-v2p-ca9.dtb -append "console=ttyAMA0
-nographic
0.000000] Booting Linux on physical CPU 0x0
0.000000] Linux version 7.1.0-rc7-ge21ee273e6fa (komal@komal-VirtualBox) (arm-linux-gnueabi-hf-gcc (Ubuntu 15.2.0-16ubuntu1) 15.2.0, GNU ld (GNU B
inutils for Ubuntu) 2.46) #1 SMP Mon Jun 15 03:26:33 +07 2026
0.000000] CPU: ARMv7 Processor [410fc090] revision 0 (ARMv7), cr=10c5387d
0.000000] CPU: PIPT / VIPT nonaliasing data cache, VIPT nonaliasing instruction cache
0.000000] OF: fdt: Machine model: V2P-CA9
0.000000] Memory policy: Data cache writeback
0.000000] efi: UEFI not found.
0.000000] cma: Failed to reserve 64 MiB
0.000000] Reserved memory: created DMA memory pool at 0x4c000000, size 8 MiB
0.000000] OF: reserved mem: initialized node vram@4c000000, compatible id shared-dma-pool
0.000000] OF: reserved mem: 0x4c000000..0x4c7fffff (8192 KiB) nomap non-reusable vram@4c000000
0.000000] CPU: All CPU(s) started in SVC mode.
0.000000] Zone ranges:

```

8) и запуск упал потому что нет корневой файловой системы

```

4.439496] 0108          65536 ram8
4.439502] (driver?)
4.439520] 0109          65536 ram9
4.439525] (driver?)
4.439543] 010a          65536 ram10
4.439548] (driver?)
4.439566] 010b          65536 ram11
4.439572] (driver?)
4.439590] 010c          65536 ram12
4.439595] (driver?)
4.439613] 010d          65536 ram13
4.439620] (driver?)
4.439638] 010e          65536 ram14
4.439643] (driver?)
4.439661] 010f          65536 ram15
4.439667] (driver?)
4.439692] 1f00          131072 mtdblock0
4.439702] (driver?)
4.439754] List of all bdev filesystems:
4.439773] ext3
4.439783] ext2
4.439794] ext4
4.439804] squashfs
4.439814] vfat
4.439824] msdos
4.439834] ntfs
4.439935]
4.440103] Kernel panic - not syncing: VFS: Unable to mount root fs on unknown-block(0,0)
4.445170] CPU: 0 UID: 0 PID: 1 Comm: swapper/0 Not tainted 7.1.0-rc7-ge21ee273e6fa #1 VOLUNTARY
4.445685] Hardware name: ARM-Versatile Express
4.446014] Call trace:
4.446543] unwind_backtrace from show_stack+0x10/0x14
4.447318] show_stack from dump_stack_lvl+0x54/0x68
4.447697] dump_stack_lvl from vpanic+0x234/0x450
4.447954] vpanic from do_panic_on_target_cpu+0x0/0x14
4.450692] ---[ end Kernel panic - not syncing: VFS: Unable to mount root fs on unknown-block(0,0) ]---

```